

■ MODULE 1

TOPIC 1: INTRODUCTION TO MACHINE LEARNING

(Mid-Sem Focused, Question-wise)

■ Q1. Define Machine Learning

Asked in: ■ Mid-Sem 2022, 2024

Answer (write as it is):

Machine Learning is a field of Artificial Intelligence that enables a computer system to learn from data and improve its performance **without being explicitly programmed**.

■ Q2. Explain Machine Learning using Tom Mitchell's definition

Asked in: ■ Mid-Sem 2023

Answer:

A computer program is said to learn from experience **E** with respect to some task **T** and performance measure **P**, if its performance at task **T**, measured by **P**, improves with experience **E**.

Example:

- **T:** Spam detection
 - **E:** Past emails
 - **P:** Accuracy
-

■ Q3. Why do we need Machine Learning?

Asked in: ■ Mid-Sem 2024

Answer (any 3 points):

1. Traditional programming fails for complex problems
2. ML systems can automatically learn patterns
3. Works efficiently with large datasets
4. Improves performance with experience

■ Q4. Machine Learning vs Traditional Programming

Asked in: ■ Mid-Sem 2023, 2024

Answer (Table – preferred):

Traditional Programming	Machine Learning
Rules are manually written	Rules are learned from data
Needs explicit logic	No explicit logic
Static system	Adaptive system
Poor scalability	Handles big data

■ Q5. List real-world applications of Machine Learning

Asked in: ■ Mid-Sem 2022

Answer:

1. Spam email filtering
 2. Recommendation systems
 3. Face recognition
 4. Medical diagnosis
 5. Stock price prediction
-

■ Q6. Identify whether the problem requires ML or not (Conceptual)

Asked in: ■ Mid-Sem 2024 (MCQ-based)

Answer logic:

ML is preferred when:

- Rules are unknown
 - Data is large
 - Problem needs prediction or pattern recognition
-

✓ END OF TOPIC 1 (Intro to ML)

You are **100% covered** for this topic in mid-sem.

■ MODULE 1

TOPIC 2: TYPES OF MACHINE LEARNING

(Supervised, Unsupervised, Semi-Supervised, Reinforcement)

■ Q1. What are the different types of Machine Learning?

Asked in: ■ Mid-Sem 2022, 2023, 2024

Answer (write as it is):

Machine Learning is broadly classified into the following types:

1. Supervised Learning
 2. Unsupervised Learning
 3. Semi-Supervised Learning
 4. Reinforcement Learning
-

■ Q2. Define Supervised Learning

Asked in: ■ Mid-Sem 2022, 2024

Answer:

Supervised Learning is a type of machine learning in which the model is trained using **labeled data**, where each input has a corresponding known output.

■ Q3. What are the tasks performed under supervised learning?

Asked in: ■ Mid-Sem 2023

Answer:

Supervised learning performs two main tasks:

1. **Regression** – predicts continuous values
 2. **Classification** – predicts discrete class labels
-

■ Q4. Give examples of supervised learning algorithms

Asked in: ■ Mid-Sem 2024

Answer:

- Linear Regression
 - Logistic Regression
 - K-Nearest Neighbour (KNN)
 - Support Vector Machine (SVM)
-

■ Q5. Define Unsupervised Learning

Asked in: ■ Mid-Sem 2022, 2023

Answer:

Unsupervised Learning is a type of machine learning where the model is trained using **unlabeled data** and learns patterns without predefined output labels.

■ Q6. What are the main objectives of unsupervised learning?

Asked in: ■ Mid-Sem 2024

Answer (any two):

1. Discover hidden patterns in data
 2. Group similar data points
 3. Reduce dimensionality
-

■ Q7. Give examples of unsupervised learning algorithms

Asked in: ■ Mid-Sem 2023

Answer:

- K-Means Clustering
 - Hierarchical Clustering
 - Principal Component Analysis (PCA)
-

■ Q8. Differentiate between Supervised and Unsupervised Learning

Asked in: ■ Mid-Sem 2023, 2024

Answer (Table – always preferred):

Supervised Learning	Unsupervised Learning
Uses labeled data	Uses unlabeled data
Output is known	Output is unknown
Prediction based	Pattern discovery based
Higher accuracy	Exploratory analysis

■ Q9. What is Semi-Supervised Learning?

Asked in: ■ Mid-Sem 2024

Answer:

Semi-Supervised Learning is a learning approach that uses a **small amount of labeled data** along with a **large amount of unlabeled data** for training.

6 of 62

■ Q10. Why is semi-supervised learning used?

Asked in: ■ Mid-Sem 2024 (short reasoning)

Answer:

1. Labeling data is expensive and time-consuming
 2. Improves accuracy compared to unsupervised learning
 3. Uses unlabeled data efficiently
-

■ Q11. What is Reinforcement Learning?

Asked in: ■ Mid-Sem 2022, 2023

Answer:

Reinforcement Learning is a learning technique where an **agent** learns by interacting with an **environment** and receives **rewards or penalties** based on its actions.

■ Q12. Name the key components of Reinforcement Learning

Asked in: ■ Mid-Sem 2023

Answer:

1. Agent
2. Environment
3. Action
4. Reward
5. Policy

■ Q13. Give examples of Reinforcement Learning

Asked in: ■ Mid-Sem 2024

Answer:

- Game playing (Chess, Ludo)
- Robotics
- Autonomous vehicles

■ Q14. Identify the type of ML used in the following problems

Asked in: ■ Mid-Sem 2024 (MCQ / short answer)

Answer:

Problem	Learning Type
Spam email detection	Supervised
Customer segmentation	Unsupervised
Game playing	Reinforcement
Image clustering	Unsupervised

■ MODULE 1

TOPIC 3: REGRESSION & LINEAR REGRESSION (MID-SEM)

■ Q1. What is Regression in Machine Learning?

Asked in: ■ Mid-Sem 2022, 2023

Answer (write exactly):

Regression is a supervised machine learning technique used to predict a **continuous-valued output** based on one or more input variables.

Examples:

- House price prediction
- Salary prediction

✂ Stop here for mid-sem

■ Q2. Define Linear Regression

Asked in: ■ Mid-Sem 2022, 2024

Answer:

Linear Regression is a supervised learning algorithm that models the relationship between an independent variable and a dependent variable using a **linear equation**.

Mathematical model:

$$y=mx+cy = mx + cy=mx+c$$

where

- mmm = slope
 - ccc = intercept
-

■ Q3. Why is Linear Regression called a supervised learning algorithm?

Asked in: ■ Mid-Sem 2023

Answer:

Linear Regression is called supervised learning because it is trained on **labeled data**, where both input values and corresponding output values are known.

■ Q4. What are the components of Linear Regression?

Asked in: ■ Mid-Sem 2024

Answer:

1. Input variable (independent variable)
 2. Output variable (dependent variable)
 3. Model parameters (slope and intercept)
 4. Cost function
-

■ Q5. What is the Cost Function in Linear Regression?

Asked in: ■ Mid-Sem 2023, 2024

Answer:

A **cost function** measures the error between the **predicted output** and the **actual output** of a linear regression model.

Commonly used cost function:

Mean Squared Error (MSE)

$$J(m,c) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

■ Q6. Why do we use Mean Squared Error (MSE)?

Asked in: ■ Mid-Sem 2024

Answer (any three):

1. Avoids cancellation of positive and negative errors
 2. Penalizes larger errors more
 3. Makes the cost function differentiable
 4. Easier optimization
-

■ Q7. What is the objective of Linear Regression?

Asked in: ■ Mid-Sem 2022

Answer:

The objective of Linear Regression is to find the **best-fitting straight line** that minimizes the **cost function (MSE)**.

■ **Q8. Write the steps involved in Linear Regression**

Asked in: ■ Mid-Sem 2023

Answer:

1. Assume a linear model
 2. Predict output values
 3. Calculate cost function
 4. Minimize the cost function
 5. Obtain optimal parameters
-

■ **Q9. What is Simple Linear Regression?**

Asked in: ■ Mid-Sem 2024

Answer:

Simple Linear Regression is a regression technique that uses **one independent variable** to predict a continuous dependent variable.

■ **Q10. Short Numerical – Linear Regression**

Asked in: ■ Mid-Sem 2023, 2024

Question pattern:

Fit a straight line for the given data.

x y

1 2

2 4

Solution (mid-sem level):

Slope:

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 - 2}{2 - 1} = 2$$

Intercept:

$$c = y - mx = 2 - 2(1) = 0$$

Final equation:

$$y = 2x$$

✓ This is sufficient for full marks

■ Q11. Linear Regression vs Logistic Regression

Asked in: ■ Mid-Sem 2024

Linear Regression	Logistic Regression
Predicts continuous output	Predicts categorical output
Uses straight line	Uses sigmoid function
Regression problem	Classification problem

⚠ What NOT to write in Mid-Sem

- Normal Equation ✗
 - Matrix form ✗
 - Partial derivatives ✗
 - Full derivations ✗
-

✓ MID-SEM SUMMARY (VERY IMPORTANT)

- **1–2 marks:** definition, cost function
- **3–5 marks:** comparison or short numerical
- Keywords examiners expect:
continuous output, linear model, MSE, labeled data

You are now **100% covered for Linear Regression (mid-sem)**.

■ MODULE 1

TOPIC 4: GRADIENT DESCENT (MID-SEM)

■ Q1. What is Gradient Descent?

Asked in: ■ Mid-Sem 2022, 2023, 2024

Answer (write exactly):

Gradient Descent is an iterative optimization algorithm used to minimize the **cost function** by updating model parameters in the direction of **negative gradient**.

■ Q2. Why is Gradient Descent used in Linear Regression?

Asked in: ■ Mid-Sem 2023, 2024

Answer:

Gradient Descent is used in Linear Regression because:

1. It efficiently minimizes the cost function
 2. Works well for large datasets
 3. Avoids computational complexity of closed-form solution
-

■ Q3. What is Learning Rate in Gradient Descent?

Asked in: ■ Mid-Sem 2022, 2024

Answer:

Learning rate (α) is a hyperparameter that controls the **step size** taken in each iteration during gradient descent.

■ Q4. Effect of Learning Rate on Gradient Descent

Asked in: ■ Mid-Sem 2023, 2024

Answer:

- **Too small α** → slow convergence
 - **Optimal α** → fast convergence
 - **Too large α** → overshooting and divergence
-

■ Q5. Write the steps of Gradient Descent

Asked in: ■ Mid-Sem 2022

Answer:

1. Initialize parameters randomly
 2. Compute cost function
 3. Compute gradient
 4. Update parameters
 5. Repeat until convergence
-

■ **Q6. What are the types of Gradient Descent?**

Asked in: ■ Mid-Sem 2023, 2024

Answer:

1. Batch Gradient Descent
 2. Stochastic Gradient Descent
 3. Mini-Batch Gradient Descent
-

■ **Q7. Define Batch Gradient Descent**

Asked in: ■ Mid-Sem 2023

Answer:

Batch Gradient Descent computes the gradient using the **entire training dataset** in each iteration.

■ **Q8. Define Stochastic Gradient Descent**

Asked in: ■ Mid-Sem 2023, 2024

Answer:

Stochastic Gradient Descent (SGD) updates parameters using **one training example at a time**.

■ **Q9. Define Mini-Batch Gradient Descent**

Asked in: ■ Mid-Sem 2024

Answer:

Mini-Batch Gradient Descent updates parameters using a **small subset of training data** in each iteration.

Q10. Compare types of Gradient Descent

Asked in: Mid-Sem 2024

Answer (Table – preferred):

Type	Data Used	Speed	Stability
Batch GD	Entire dataset	Slow	Stable
Stochastic GD	Single data point	Fast	Noisy
Mini-batch GD	Small batch	Balanced	Stable

Q11. What are the advantages of Gradient Descent?

Asked in: Mid-Sem 2024

Answer (any three):

1. Easy to implement
 2. Works for large datasets
 3. Scales well with features
 4. Efficient optimization
-

Q12. What are the limitations of Gradient Descent?

Asked in: Mid-Sem 2023

Answer (any two):

1. Sensitive to learning rate
2. May converge to local minima
3. Slow convergence for small α

MODULE 1

TOPIC 5: MULTIPLE LINEAR REGRESSION (MID-SEM)

■ Q1. What is Multiple Linear Regression?

Asked in: ■ Mid-Sem 2023, 2024

Answer (write exactly):

Multiple Linear Regression (MLR) is a supervised learning algorithm that models the relationship between **two or more independent variables** and a **continuous dependent variable** using a linear equation.

■ Q2. Write the mathematical model of Multiple Linear Regression

Asked in: ■ Mid-Sem 2024

Answer:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

where:

- y = dependent variable
 - $x_1, x_2, \dots, x_n$ = independent variables
 - β_0 = intercept
 - $\beta_1, \beta_2, \dots$ = coefficients
-

■ Q3. How is Multiple Linear Regression different from Simple Linear Regression?

Asked in: ■ Mid-Sem 2023

Answer (Table – preferred):

Simple Linear Regression	Multiple Linear Regression
One independent variable	Two or more independent variables
Model: $y = mx + c$	Model: $y = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n$
Easy to visualize	Difficult to visualize

■ Q4. Give examples of Multiple Linear Regression

Asked in: ■ Mid-Sem 2024

Answer:

- Predicting house price using area, number of rooms, location
 - Salary prediction using experience, education, skills
-

■ Q5. What is multicollinearity?

Asked in: ■ Mid-Sem 2024

Answer:

Multicollinearity occurs when two or more independent variables in a multiple linear regression model are **highly correlated** with each other.

■ Q6. Why is multicollinearity a problem?

Asked in: ■ Mid-Sem 2024

Answer (any two):

1. Makes coefficient estimates unstable
 2. Reduces interpretability of the model
 3. Increases variance of coefficients
-

■ Q7. Short Numerical – Multiple Linear Regression

Asked in: ■ Mid-Sem 2023 (Make-up)

Question pattern:

Given the regression equation, predict the output.

$$y = 2 + 3x_1 + 4x_2 \quad y = 2 + 3x_1 + 4x_2$$

$$\text{For } x_1 = 1, x_2 = 2 \quad x_1 = 1, x_2 = 2$$

Solution:

$$y = 2 + 3(1) + 4(2) = 2 + 3 + 8 = 13 \quad y = 2 + 3(1) + 4(2) = 2 + 3 + 8 = 13$$

✓ Enough for full marks

■ Q8. List assumptions of Multiple Linear Regression

Asked in: ■ Mid-Sem 2024

Answer (any four):

1. Linear relationship between variables
2. No multicollinearity
3. Errors are independent
4. Constant variance of errors
5. Normally distributed errors

 **Do not explain unless asked**

 **What NOT to write in Mid-Sem**

- Matrix form 
 - Normal Equation 
 - Derivations 
-

 **MID-SEM SUMMARY**

- Definition → **1–2 marks**
- Model + comparison → **3–5 marks**
- Short numerical → **very common**
- Keywords examiners expect:
multiple inputs, linear model, multicollinearity

You are now **fully covered for Multiple Linear Regression (mid-sem)**.

MODULE 1

TOPIC 6: CLOSED-FORM EQUATION & TYPES OF GRADIENT DESCENT (MID-SEM)

 **Q1. What is a Closed-Form Equation in Linear Regression?**

Asked in:  Mid-Sem 2023, 2024

Answer (write exactly):

A **Closed-Form Equation** is a direct mathematical solution used to compute the optimal parameters of a linear regression model **without using iterations**.

Q2. Write the closed-form (normal) equation for Linear Regression

Asked in: Mid-Sem 2024

Answer:

$$\theta = (X^T X)^{-1} X^T y$$

⚠ Do NOT derive this in mid-sem

Q3. Why is the closed-form equation not preferred for large datasets?

Asked in: Mid-Sem 2023

Answer (any two):

1. Matrix inversion is computationally expensive
 2. High memory requirement
 3. Not scalable for large datasets
-

Q4. Closed-Form Equation vs Gradient Descent

Asked in: Mid-Sem 2024

Answer (Table – preferred):

Closed-Form Equation	Gradient Descent
Direct solution	Iterative solution
No learning rate	Requires learning rate
Expensive for large data	Efficient for large data
One-time computation	Multiple iterations

Q5. When should Gradient Descent be preferred over Closed-Form Equation?

Asked in: Mid-Sem 2024

Answer:

Gradient Descent is preferred when:

1. Dataset is very large

2. Number of features is high
 3. Memory constraints exist
-

■ **Q6. List the types of Gradient Descent**

Asked in: ■ Mid-Sem 2023, 2024

Answer:

1. Batch Gradient Descent
 2. Stochastic Gradient Descent
 3. Mini-Batch Gradient Descent
-

■ **Q7. What is Batch Gradient Descent?**

Asked in: ■ Mid-Sem 2023

Answer:

Batch Gradient Descent updates model parameters using the **entire training dataset** in each iteration.

■ **Q8. What is Stochastic Gradient Descent?**

Asked in: ■ Mid-Sem 2024

Answer:

Stochastic Gradient Descent updates model parameters using **one training example at a time**.

■ **Q9. What is Mini-Batch Gradient Descent?**

Asked in: ■ Mid-Sem 2024

Answer:

Mini-Batch Gradient Descent updates parameters using a **small subset of training data** in each iteration.

■ **Q10. Compare Batch, Stochastic and Mini-Batch Gradient Descent**

Asked in:  Mid-Sem 2024

Answer (Table – preferred):

Type	Data Used	Speed	Stability
Batch GD	Full dataset	Slow	Stable
SGD	Single sample	Fast	Noisy
Mini-Batch GD	Small batch	Balanced	Stable

 **What NOT to write in Mid-Sem**

- Proofs 
 - Matrix derivations 
 - Convergence theory 
-

 **MID-SEM SUMMARY**

- Definition + formula → **1–2 marks**
 - Comparison table → **3–5 marks**
 - Preference-based questions → **very common**
-

MODULE 1

TOPIC 7: NORMALIZATION, STANDARDIZATION, OVERFITTING & UNDERFITTING (MID-SEM)

 **Q11. What is Normalization?**

Asked in:  Mid-Sem 2023, 2024

Answer:

Normalization is a data preprocessing technique used to scale feature values into a **fixed range**, usually between 0 and 1.

Formula (Min-Max):

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

 **Q12. What is Standardization?**

Asked in: ■ Mid-Sem 2024

Answer:

Standardization scales data such that it has **zero mean and unit variance**.

Formula (Z-score):

$$x' = \frac{x - \mu}{\sigma} \quad x' = \sigma x - \mu$$

■ **Q13. Differentiate between Normalization and Standardization**

Asked in: ■ Mid-Sem 2023, 2024

Normalization	Standardization
Scales to fixed range	Centers around mean
Sensitive to outliers	Handles outliers better
Uses min & max	Uses mean & std dev

■ **Q14. Why is feature scaling important?**

Asked in: ■ Mid-Sem 2024

Answer (any three):

1. Improves model convergence
 2. Required for distance-based algorithms (KNN)
 3. Prevents dominance of large-scale features
-

■ **Q15. What is Overfitting?**

Asked in: ■ Mid-Sem 2022, 2023

Answer:

Overfitting occurs when a model performs very well on training data but poorly on unseen test data.

■ **Q16. What is Underfitting?**

Asked in: ■ Mid-Sem 2023

Answer:

Underfitting occurs when a model is too simple to capture the underlying pattern in the data.

■ Q17. Overfitting vs Underfitting

Asked in: ■ Mid-Sem 2024

Overfitting	Underfitting
High variance	High bias
Complex model	Simple model
Poor generalization	Poor learning

⚠ **Stop here for mid-sem**

(No curves, no proofs)

✅ MODULE 1 STATUS

- ✓ Introduction
- ✓ Types of ML
- ✓ Linear Regression
- ✓ Gradient Descent
- ✓ Multiple Linear Regression
- ✓ Closed-Form Equation
- ✓ Normalization & Overfitting

■ MODULE 1

TOPIC 8: BIAS, VARIANCE & BIAS–VARIANCE TRADE-OFF (MID-SEM)

■ Q1. What is Bias in Machine Learning?

Asked in: ■ Mid-Sem 2023, 2024

Answer (write exactly):

Bias is the error introduced by simplifying a real-world problem using an overly simple model.

■ Q2. What is Variance in Machine Learning?

Asked in: ■ Mid-Sem 2023, 2024

Answer:

Variance is the error caused by a model being too sensitive to small changes in the training data.

■ **Q3. How are bias and variance related to underfitting and overfitting?**

Asked in: ■ Mid-Sem 2024

Answer:

- **High bias** → Underfitting
 - **High variance** → Overfitting
-

■ **Q4. What is Bias–Variance Trade-off?**

Asked in: ■ Mid-Sem 2023, 2024

Answer:

The **bias–variance trade-off** refers to the balance between bias and variance that must be achieved to minimize total model error.

■ **Q5. Why is bias–variance trade-off important?**

Asked in: ■ Mid-Sem 2024

Answer (any three):

1. Helps improve model generalization
 2. Prevents overfitting and underfitting
 3. Achieves optimal model complexity
-

■ **Q6. How can overfitting be reduced?**

Asked in: ■ Mid-Sem 2024

Answer (any four):

1. Increase training data
2. Reduce model complexity

3. Apply regularization
4. Feature selection

■ MODULE 1

TOPIC 9: REGULARIZATION (RIDGE & LASSO) (MID-SEM)

■ Q7. What is Regularization?

Asked in: ■ Mid-Sem 2023, 2024

Answer (write exactly):

Regularization is a technique used to prevent overfitting by adding a **penalty term** to the cost function.

■ Q8. Why is Regularization needed?

Asked in: ■ Mid-Sem 2024

Answer (any three):

1. Reduces overfitting
 2. Controls model complexity
 3. Improves generalization
-

■ Q9. What is Ridge Regression?

Asked in: ■ Mid-Sem 2023, 2024

Answer:

Ridge Regression is a regularization technique that adds an **L2 penalty** (sum of squares of coefficients) to the cost function.

■ Q10. What is Lasso Regression?

Asked in: ■ Mid-Sem 2024

Answer:

Lasso Regression is a regularization technique that adds an **L1 penalty** (sum of absolute values of coefficients) to the cost function.

■ Q11. Difference between Ridge and Lasso Regression

Asked in: ■ Mid-Sem 2024

Ridge Regression	Lasso Regression
Uses L2 penalty	Uses L1 penalty
Shrinks coefficients	Can make coefficients zero
No feature selection	Performs feature selection

■ Q12. Effect of regularization parameter (λ)

Asked in: ■ Mid-Sem 2024

Answer:

- Large $\lambda \rightarrow$ more penalty $\rightarrow$ simpler model
- Small $\lambda \rightarrow$ less penalty $\rightarrow$ complex model

✓ MODULE 1 COMPLETION STATUS

- ✓ Intro to ML
- ✓ Types of ML
- ✓ Linear Regression
- ✓ Gradient Descent
- ✓ Multiple Linear Regression
- ✓ Closed-Form Equation
- ✓ Normalization & Overfitting
- ✓ Bias-Variance
- ✓ Regularization

👉 MODULE 1 = 100% DONE (Mid-Sem) 🎉

■ MODULE 2

TOPIC 10: CLASSIFICATION & LOGISTIC REGRESSION – PART 1 (MID-SEM)

■ Q1. What is Classification in Machine Learning?

Asked in: ■ Mid-Sem 2022, 2023, 2024

Answer (write exactly):

Classification is a supervised machine learning task in which the model predicts discrete class labels for given input data.

Examples:

- Spam vs non-spam email
 - Disease: positive / negative
-

■ Q2. Differentiate between Regression and Classification

Asked in: ■ Mid-Sem 2023, 2024

Answer (Table – preferred):

Regression	Classification
Predicts continuous output	Predicts discrete classes
Example: house price	Example: spam detection
Output is numeric	Output is categorical

■ Q3. Why can't Linear Regression be used for Classification?

Asked in: ■ Mid-Sem 2024

Answer:

Linear Regression is not suitable for classification because:

1. Output is unbounded (can go beyond 0 and 1)
 2. Cannot model probabilities
 3. Sensitive to outliers
-

■ Q4. What is Logistic Regression?

Asked in: ■ Mid-Sem 2022, 2023, 2024

Answer (write exactly):

Logistic Regression is a supervised learning algorithm used for binary classification, where the output is modeled using a sigmoid function.

■ Q5. What is Binary Classification?

Asked in: ■ Mid-Sem 2023

Answer:

Binary classification is a classification problem in which the output belongs to one of two possible classes, such as 0 or 1.

■ Q6. What is the Sigmoid Function?

Asked in: ■ Mid-Sem 2023, 2024

Answer:

The sigmoid function maps any real-valued input into a range between 0 and 1, making it suitable for probability estimation.

Formula:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

■ Q7. Why is Sigmoid Function used in Logistic Regression?

Asked in: ■ Mid-Sem 2024

Answer (any three):

1. Output lies between 0 and 1
 2. Can be interpreted as probability
 3. Smooth and differentiable
-

■ Q8. What is the decision boundary in Logistic Regression?

Asked in: ■ Mid-Sem 2024

Answer:

The decision boundary is a threshold (usually 0.5) that separates different classes based on predicted probabilities.

■ Q9. Write the hypothesis function of Logistic Regression

Asked in: ■ Mid-Sem 2023

Answer:

$$h(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}} \quad h(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

⚠ No derivation needed for mid-sem

■ Q10. Logistic Regression vs Linear Regression

Asked in: ■ Mid-Sem 2024

Linear Regression	Logistic Regression
Continuous output	Binary output
Uses straight line	Uses sigmoid curve
Least squares loss	Log loss

■ MODULE 2

TOPIC 11: LOGISTIC REGRESSION – PART 2 (MID-SEM)

■ Q1. What is the cost (loss) function used in Logistic Regression?

Asked in: ■ Mid-Sem 2023, 2024

Answer (write exactly):

Logistic Regression uses the Log Loss (Binary Cross-Entropy) cost function to measure the difference between actual and predicted class probabilities.

Formula:

$$J(y, \hat{y}) = -[y \log(\hat{y}) + (1-y) \log(1-\hat{y})] \quad J(y, \hat{y}) = -[y \log(\hat{y}) + (1-y) \log(1-\hat{y})]$$

⚠ No derivation needed for mid-sem

■ Q2. Why is Mean Squared Error not used in Logistic Regression?

Asked in: ■ Mid-Sem 2024

Answer:

Mean Squared Error is not used in Logistic Regression because:

1. It leads to a non-convex cost function
2. Causes slow or incorrect convergence

3. Not suitable for probability outputs

■ Q3. What is Log Loss?

Asked in: ■ Mid-Sem 2023

Answer:

Log Loss measures how close the predicted probability is to the actual class label and penalizes confident wrong predictions heavily.

■ Q4. What do you mean by odds and log-odds in Logistic Regression?

Asked in: ■ Mid-Sem 2024

Answer:

- Odds = Probability of success / Probability of failure
- Log-odds (logit) = logarithm of odds

$$\text{logit}(p) = \log\left(\frac{p}{1-p}\right) \quad \text{logit}(p) = \log\left(\frac{p}{1-p}\right)$$

■ Q5. Why is Logistic Regression called a linear classifier?

Asked in: ■ Mid-Sem 2024

Answer:

Logistic Regression is called a linear classifier because its decision boundary is a linear function of the input features.

■ Q6. What are the advantages of Logistic Regression?

Asked in: ■ Mid-Sem 2023

Answer (any three):

1. Simple and efficient
2. Outputs probabilities
3. Easy to interpret
4. Less prone to overfitting

■ Q7. What are the limitations of Logistic Regression?

Asked in: ■ Mid-Sem 2024

Answer (any three):

1. Works only for linear decision boundaries
2. Not suitable for complex relationships
3. Sensitive to outliers

■ Q8. What is regularization in Logistic Regression?

Asked in: ■ Mid-Sem 2024

Answer:

Regularization in Logistic Regression is used to prevent overfitting by adding a penalty term to the cost function.

■ Q9. What type of regularization is used in Logistic Regression?

Asked in: ■ Mid-Sem 2024

Answer:

- L1 Regularization (Lasso)
- L2 Regularization (Ridge)

■ MODULE 2

TOPIC 12: NEAREST NEIGHBOUR & K-NEAREST NEIGHBOUR (KNN) (MID-SEM)

■ Q1. What is Nearest Neighbour Classification?

Asked in: ■ Mid-Sem 2022, 2023

Answer (write exactly):

Nearest Neighbour Classification assigns a class to a test sample based on the class of the closest training data point.

■ Q2. What is K-Nearest Neighbour (KNN)?

Asked in: ■ Mid-Sem 2022, 2023, 2024

Answer:

K-Nearest Neighbour (KNN) is a supervised learning algorithm that classifies a data point based on the majority class of its K nearest neighbors.

■ Q3. Is KNN a lazy learner? Why?

Asked in: ■ Mid-Sem 2023, 2024

Answer:

Yes, KNN is called a lazy learner because:

1. It does not learn a model during training
 2. All computation is done during prediction time
-

■ Q4. Write the steps of KNN algorithm

Asked in: ■ Mid-Sem 2023

Answer:

1. Choose the value of K
 2. Calculate distance between test point and training points
 3. Select K nearest neighbors
 4. Assign class by majority voting
-

■ Q5. What distance metrics are used in KNN?

Asked in: ■ Mid-Sem 2024

Answer:

- Euclidean distance
 - Manhattan distance
 - Minkowski distance
-

■ Q6. Effect of value of K in KNN

Asked in: ■ Mid-Sem 2023, 2024

Answer:

- Small K → low bias, high variance
 - Large K → high bias, low variance
-

■ Q7. Why is feature scaling important in KNN?

Asked in: ■ Mid-Sem 2024

Answer:

Feature scaling is important in KNN because distance calculations can be dominated by features with larger scales.

■ Q8. Advantages of KNN

Asked in: ■ Mid-Sem 2023

Answer (any three):

1. Simple and easy to understand
 2. No training phase
 3. Effective for small datasets
-

■ Q9. Limitations of KNN

Asked in: ■ Mid-Sem 2024

Answer (any three):

1. Computationally expensive
 2. Sensitive to noise
 3. Requires feature scaling
-

■ Q10. Short Numerical – KNN Classification

Asked in:  Mid-Sem 2023, 2024

Question type:

Classify a test point using KNN.

Training data:

Point	(x, y)	Class
A	(1,1)	Red
B	(2,2)	Red
C	(4,4)	Blue

Test point: (3,3), $K = 3$

Solution:

Distances:

- $d(A) = \sqrt{((3-1)^2 + (3-1)^2)} = \sqrt{8}$
- $d(B) = \sqrt{((3-2)^2 + (3-2)^2)} = \sqrt{2}$
- $d(C) = \sqrt{((3-4)^2 + (3-4)^2)} = \sqrt{2}$

Nearest neighbors: B, C, A
Majority class = Red

Predicted class: Red

Q11. KNN vs Logistic Regression

Asked in:  Mid-Sem 2024

KNN	Logistic Regression
Instance-based	Model-based
No training	Training required
Non-parametric	Parametric

MODULE 2

TOPIC 13: ERROR ANALYSIS & PERFORMANCE METRICS (MID-SEM)

■ Q1. What is Error Analysis in Machine Learning?

Asked in: ■ Mid-Sem 2023, 2024

Answer (write exactly):

Error Analysis is the process of evaluating a machine learning model by analyzing the difference between predicted outputs and actual outputs.

■ Q2. What is Train–Test Split?

Asked in: ■ Mid-Sem 2022, 2023, 2024

Answer:

Train–Test Split is a technique in which the dataset is divided into a training set and a testing set to evaluate model performance.

Typical split:

- **70% training**
 - **30% testing**
-

■ Q3. Why do we use a validation set?

Asked in: ■ Mid-Sem 2024

Answer:

A validation set is used to:

- 1. Tune hyperparameters**
 - 2. Prevent overfitting**
 - 3. Select the best model**
-

■ Q4. What is a Confusion Matrix?

Asked in: ■ Mid-Sem 2022, 2023, 2024

Answer:

A Confusion Matrix is a table used to describe the performance of a classification model by comparing actual and predicted class labels.

Structure:

Actual \ Predicted	Positive	Negative
Positive	TP	FN
Negative	FP	TN

■ Q5. Define Accuracy

Asked in: ■ Mid-Sem 2022, 2023, 2024

Answer:

Accuracy is the ratio of correctly predicted observations to the total observations.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

■ Q6. Define Precision

Asked in: ■ Mid-Sem 2023, 2024

Answer:

Precision is the ratio of correctly predicted positive observations to the total predicted positive observations.

$$\text{Precision} = \frac{TP}{TP + FP}$$

■ Q7. Define Recall

Asked in: ■ Mid-Sem 2023, 2024

Answer:

Recall is the ratio of correctly predicted positive observations to all actual positive observations.

$$\text{Recall} = \frac{TP}{TP + FN}$$

■ Q8. What is F1-Score?

Asked in: ■ Mid-Sem 2024

Answer:

F1-Score is the harmonic mean of Precision and Recall.

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

■ Q9. When is Precision more important than Recall?

Asked in: ■ Mid-Sem 2024

Answer:

Precision is more important when false positives are costly, such as:

- Spam email filtering
 - Fraud detection
-

■ Q10. When is Recall more important than Precision?

Asked in: ■ Mid-Sem 2024

Answer:

Recall is more important when false negatives are costly, such as:

- Disease detection
 - Security threats
-

■ Q11. What is ROC Curve?

Asked in: ■ Mid-Sem 2024

Answer:

ROC (Receiver Operating Characteristic) Curve is a graphical plot that shows the trade-off between True Positive Rate and False Positive Rate.

■ Q12. What does AUC represent?

Asked in: ■ Mid-Sem 2024

Answer:

AUC (Area Under Curve) represents the overall performance of a classification model. Higher AUC indicates better model performance.

■ Q13. Short Numerical – Performance Metrics

Asked in: ■ Mid-Sem 2023, 2024

Given:

- TP = 40
- TN = 50
- FP = 10
- FN = 5

Solution:

Accuracy:

$$\frac{40+50}{40+50+10+5} = \frac{90}{105} = 0.857$$

Precision:

$$\frac{40}{40+10} = 0.8$$

Recall:

$$\frac{40}{40+5} = 0.89$$

MODULE 2

TOPIC 14: NAIVE BAYES CLASSIFIER – PART 1 (MID-SEM)

Q1. What is Naive Bayes Classifier?

Asked in: Mid-Sem 2022, 2023, 2024

Answer (write exactly):

Naive Bayes Classifier is a supervised probabilistic machine learning algorithm based on Bayes' Theorem, with the assumption that features are conditionally independent.

Q2. State Bayes' Theorem

Asked in: Mid-Sem 2022, 2023

Answer:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

where:

- $P(A|B)$ = Posterior probability

- $P(B|A)P(B|A)P(B|A)$ = Likelihood
 - $P(A)P(A)P(A)$ = Prior probability
 - $P(B)P(B)P(B)$ = Evidence
-

■ Q3. Why is Naive Bayes called “Naive”?

Asked in: ■ Mid-Sem 2023, 2024

Answer:

Naive Bayes is called naive because it assumes that all features are independent of each other, which is rarely true in real-world data.

■ Q4. What is the main assumption of Naive Bayes?

Asked in: ■ Mid-Sem 2024

Answer:

The main assumption of Naive Bayes is conditional independence of features given the class label.

■ Q5. Write the classification rule for Naive Bayes

Asked in: ■ Mid-Sem 2023

Answer:

A data point is assigned to the class with the maximum posterior probability.

$\text{Class} = \arg\max P(C_i | X)$

■ Q6. What are the types of Naive Bayes classifiers?

Asked in: ■ Mid-Sem 2024

Answer:

1. Gaussian Naive Bayes
 2. Multinomial Naive Bayes
 3. Bernoulli Naive Bayes
-

■ Q7. Where is Naive Bayes commonly used?

Asked in: ■ Mid-Sem 2023

Answer (any three):

1. Spam email classification
 2. Text classification
 3. Sentiment analysis
 4. Document categorization
-

■ Q8. Advantages of Naive Bayes Classifier

Asked in: ■ Mid-Sem 2023, 2024

Answer (any four):

1. Simple and easy to implement
 2. Works well with small datasets
 3. Fast training and prediction
 4. Handles high-dimensional data
-

■ Q9. Limitations of Naive Bayes Classifier

Asked in: ■ Mid-Sem 2024

Answer (any three):

1. Strong independence assumption
 2. Poor performance when features are correlated
 3. Zero-frequency problem
-

■ Q10. What is the Zero-Frequency Problem?

Asked in: ■ Mid-Sem 2024

Answer:

The zero-frequency problem occurs when a feature value never appears with a class in the training data, resulting in zero probability.

⚠ What NOT to write in Mid-Sem

- Laplace smoothing derivation ✗
 - Log probability derivation ✗
 - Continuous PDF math ✗
-

✅ MID-SEM SUMMARY

- Definition + assumption → guaranteed
- Bayes theorem → direct question
- Advantages / limitations → easy marks

You are now fully covered for Naive Bayes – Part 1 (mid-sem).

📘 MODULE 2

TOPIC 14: NAIVE BAYES CLASSIFIER – PART 1 (MID-SEM)

📘 Q1. What is Naive Bayes Classifier?

Asked in: 📘 Mid-Sem 2022, 2023, 2024

Answer (write exactly):

Naive Bayes Classifier is a supervised probabilistic machine learning algorithm based on Bayes' Theorem, with the assumption that features are conditionally independent.

📘 Q2. State Bayes' Theorem

Asked in: 📘 Mid-Sem 2022, 2023

Answer:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} \quad P(A|B)P(B) = P(B|A)P(A)$$

where:

- $P(A|B)P(B)$ = Posterior probability
- $P(B|A)P(A)$ = Likelihood

- $P(A)P(A)P(A)$ = Prior probability
 - $P(B)P(B)P(B)$ = Evidence
-

■ Q3. Why is Naive Bayes called “Naive”?

Asked in: ■ Mid-Sem 2023, 2024

Answer:

Naive Bayes is called naive because it assumes that all features are independent of each other, which is rarely true in real-world data.

■ Q4. What is the main assumption of Naive Bayes?

Asked in: ■ Mid-Sem 2024

Answer:

The main assumption of Naive Bayes is conditional independence of features given the class label.

■ Q5. Write the classification rule for Naive Bayes

Asked in: ■ Mid-Sem 2023

Answer:

A data point is assigned to the class with the maximum posterior probability.

$\text{Class} = \arg\max P(C_i | X) \text{ } \text{Class} = \arg\max P(C_i | X)$

■ Q6. What are the types of Naive Bayes classifiers?

Asked in: ■ Mid-Sem 2024

Answer:

1. Gaussian Naive Bayes
 2. Multinomial Naive Bayes
 3. Bernoulli Naive Bayes
-

■ Q7. Where is Naive Bayes commonly used?

Asked in: ■ Mid-Sem 2023

Answer (any three):

1. Spam email classification
 2. Text classification
 3. Sentiment analysis
 4. Document categorization
-

■ Q8. Advantages of Naive Bayes Classifier

Asked in: ■ Mid-Sem 2023, 2024

Answer (any four):

1. Simple and easy to implement
 2. Works well with small datasets
 3. Fast training and prediction
 4. Handles high-dimensional data
-

■ Q9. Limitations of Naive Bayes Classifier

Asked in: ■ Mid-Sem 2024

Answer (any three):

1. Strong independence assumption
 2. Poor performance when features are correlated
 3. Zero-frequency problem
-

■ Q10. What is the Zero-Frequency Problem?

Asked in: ■ Mid-Sem 2024

Answer:

The zero-frequency problem occurs when a feature value never appears with a class in the training data, resulting in zero probability.

⚠️ What NOT to write in Mid-Sem

- Laplace smoothing derivation ✗
 - Log probability derivation ✗
 - Continuous PDF math ✗
-

✅ MID-SEM SUMMARY

- Definition + assumption → guaranteed
- Bayes theorem → direct question
- Advantages / limitations → easy marks

You are now fully covered for Naive Bayes – Part 1 (mid-sem).

📘 MODULE 2

TOPIC 16: DECISION TREE – INTRODUCTION & ID3 (PART 1) — MID-SEM

📘 Q1. What is a Decision Tree?

Asked in: 📘 Mid-Sem 2022, 2023, 2024

Answer (write exactly):

A Decision Tree is a supervised machine learning algorithm used for classification and regression, where decisions are made by splitting data based on feature values in a tree-like structure.

📘 Q2. What are the main components of a Decision Tree?

Asked in: 📘 Mid-Sem 2023, 2024

Answer:

1. Root Node – topmost node
 2. Decision Nodes – internal nodes
 3. Leaf Nodes – output or class label
 4. Branches – decision rules
-

■ Q3. Why are Decision Trees popular?

Asked in: ■ Mid-Sem 2024

Answer (any three):

1. Easy to understand and interpret
 2. No need for feature scaling
 3. Handles both numerical and categorical data
-

■ Q4. What is the ID3 Algorithm?

Asked in: ■ Mid-Sem 2022, 2023, 2024

Answer:

ID3 (Iterative Dichotomiser 3) is a decision tree algorithm that constructs a tree by selecting the attribute with maximum Information Gain at each step.

■ Q5. What criterion is used by ID3 to split data?

Asked in: ■ Mid-Sem 2023

Answer:

ID3 uses Information Gain as the splitting criterion.

■ Q6. What is Entropy?

Asked in: ■ Mid-Sem 2022, 2023, 2024

Answer:

Entropy is a measure of impurity or randomness in a dataset.

Formula:

$Entropy(S) = -\sum p_i \log_2 p_i$

■ Q7. What is Information Gain?

Asked in: ■ Mid-Sem 2022, 2023, 2024

Answer:

Information Gain measures the reduction in entropy after splitting the dataset on an attribute.

Formula:

$$IG(S,A)=Entropy(S)-\sum \frac{|S_v|}{|S|} Entropy(S_v) \quad IG(S, A) = Entropy(S) - \sum \frac{|S_v|}{|S|} Entropy(S_v)$$

■ Q8. Why is Information Gain used in ID3?

Asked in: ■ Mid-Sem 2024

Answer:

Information Gain is used because it:

1. Selects the most informative attribute
 2. Reduces uncertainty efficiently
 3. Improves classification accuracy
-

■ Q9. Short Numerical – Entropy Calculation

Asked in: ■ Mid-Sem 2023, 2024

Given a dataset with:

- 6 Positive samples
- 4 Negative samples

Solution:

$$Entropy = -\left(\frac{6}{10} \log_2 \frac{6}{10} + \frac{4}{10} \log_2 \frac{4}{10}\right) \quad Entropy = -(106 \log_2 106 + 104 \log_2 104)$$

(Just substitute values — calculation not required unless asked)

■ Q10. How does ID3 select the root node?

Asked in: ■ Mid-Sem 2024

Answer:

ID3 selects the attribute with the highest Information Gain as the root node.

■ MODULE 2

TOPIC 17: DECISION TREE – OVERFITTING & PRUNING (PART 2) (MID-SEM)

■ Q1. What is Overfitting in Decision Trees?

Asked in: ■ Mid-Sem 2023, 2024

Answer (write exactly):

Overfitting in Decision Trees occurs when the tree becomes too complex and fits the training data very well but performs poorly on unseen data.

■ Q2. Why are Decision Trees prone to overfitting?

Asked in: ■ Mid-Sem 2024

Answer (any three):

1. Deep tree structure
 2. Too many splits
 3. Learning noise from training data
-

■ Q3. How can overfitting in Decision Trees be reduced?

Asked in: ■ Mid-Sem 2023, 2024

Answer:

Overfitting can be reduced by using pruning techniques:

1. Pre-pruning
 2. Post-pruning
-

■ Q4. What is Pre-Pruning?

Asked in: ■ Mid-Sem 2024

Answer:

Pre-pruning is a technique where the growth of the decision tree is stopped early by setting conditions such as maximum depth or minimum samples.

■ Q5. What is Post-Pruning?

Asked in: ■ Mid-Sem 2023, 2024

Answer:

Post-pruning is a technique where a fully grown decision tree is pruned back by removing branches that do not improve performance on validation data.

■ Q6. Differentiate between Pre-Pruning and Post-Pruning

Asked in: ■ Mid-Sem 2024

Pre-Pruning	Post-Pruning
Stops tree early	Tree grown fully first
Faster	More accurate
Risk of underfitting	Reduces overfitting better

■ Q7. Give advantages of Decision Trees

Asked in: ■ Mid-Sem 2023

Answer (any three):

1. Easy to interpret
 2. No feature scaling required
 3. Handles non-linear data
-

■ Q8. Give limitations of Decision Trees

Asked in: ■ Mid-Sem 2024

Answer (any three):

1. Prone to overfitting
2. Sensitive to small data changes
3. Can create complex trees

■ MODULE 2

TOPIC 17: DECISION TREE – OVERFITTING & PRUNING (PART 2) (MID-SEM)

■ Q1. What is Overfitting in Decision Trees?

Asked in: ■ Mid-Sem 2023, 2024

Answer (write exactly):

Overfitting in Decision Trees occurs when the tree becomes too complex and fits the training data very well but performs poorly on unseen data.

■ Q2. Why are Decision Trees prone to overfitting?

Asked in: ■ Mid-Sem 2024

Answer (any three):

1. Deep tree structure
 2. Too many splits
 3. Learning noise from training data
-

■ Q3. How can overfitting in Decision Trees be reduced?

Asked in: ■ Mid-Sem 2023, 2024

Answer:

Overfitting can be reduced by using pruning techniques:

1. Pre-pruning
 2. Post-pruning
-

■ Q4. What is Pre-Pruning?

Asked in: ■ Mid-Sem 2024

Answer:

Pre-pruning is a technique where the growth of the decision tree is stopped early by setting conditions such as maximum depth or minimum samples.

■ Q5. What is Post-Pruning?

Asked in: ■ Mid-Sem 2023, 2024

Answer:

Post-pruning is a technique where a fully grown decision tree is pruned back by removing branches that do not improve performance on validation data.

■ Q6. Differentiate between Pre-Pruning and Post-Pruning

Asked in: ■ Mid-Sem 2024

Pre-Pruning	Post-Pruning
Stops tree early	Tree grown fully first
Faster	More accurate
Risk of underfitting	Reduces overfitting better

■ Q7. Give advantages of Decision Trees

Asked in: ■ Mid-Sem 2023

Answer (any three):

1. Easy to interpret
 2. No feature scaling required
 3. Handles non-linear data
-

■ Q8. Give limitations of Decision Trees

Asked in: ■ Mid-Sem 2024

Answer (any three):

1. Prone to overfitting
2. Sensitive to small data changes
3. Can create complex trees

MODULE 1 – FOUNDATIONS & REGRESSION

Topic	Importance	Typical Mid-Sem Questions	
Introduction to ML	★ ★	Definition, ML vs Traditional	
Types of Machine Learning	★ ★ ★	Supervised vs Unsupervised	
Linear Regression	★ ★ ★ ★	Definition, cost function, numerical	
Gradient Descent	★ ★ ★ ★	Learning rate, GD types	
Multiple Linear Regression	★ ★ ★	Model, short numerical	
Closed-Form Equation	★ ★	Formula, GD vs Normal Equation	
Normalization & Standardization	★ ★ ★ ★	Formula, comparison	
Overfitting & Underfitting	★ ★ ★	Definitions, comparison	
Bias-Variance Trade-off	★ ★ ★	Conceptual explanation	
Regularization (Ridge, Lasso)	★ ★ ★ ★	Difference, λ effect	

MODULE 2 – CLASSIFICATION & MODELS

Topic	Importance	Typical Mid-Sem Questions
Classification Basics	★ ★	Regression vs Classification
Logistic Regression (Binary)	★ ★ ★ ★	Sigmoid, comparison, log loss
K-Nearest Neighbour (KNN)	★ ★ ★ ★	Algorithm, effect of K, numerical
Error Analysis & Metrics	★ ★ ★ ★	Confusion matrix, precision/recall
Naive Bayes Classifier	★ ★ ★	Assumption, Bayes theorem, numerical
Decision Tree (ID3)	★ ★ ★	Entropy, Information Gain
Pruning in Decision Trees	★ ★	Pre vs Post pruning
Support Vector Machine (SVM)	★ ★ ★	Margin, kernel, soft vs hard